HRIDAM ROSHAN PAUL

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Academic Credentials

Bachelor of Science in Mechanical Engineering **Jan 2023 – May 2027 (Expected)**
Khulna University of Engineering & Technology (KUET)

Research Interests

- Additive Manufacturing
- Fused Filament Fabrication
- Functionally Graded Materials and Structures
- Sandwich and Architected Structures
- Natural Fiber Composites
- Metal Matrix Composites
- Computational Mechanics and Finite Element Analysis
- Bio-inspired Structural Design and Topology Optimization

Research Experience

Undergraduate Thesis (Ongoing):

- **Topic:** Experimental Investigation of Graphite/PLA Composites: Fused Filament Fabrication and FGAM Performance Characterization
  Supervisor: Dr. Md. Shariful Islam, Department of Mechanical Engineering, KUET
  Summary: Planned experimental investigation of graphite-filled PLA, covering filament fabrication, fused filament fabrication (FFF) printing, and performance characterization of functionally graded additive manufactured (FGAM) specimens. The work examines how graphite loading affects thermal dissipation and interlayer mechanical integrity, using uniform, step-gradient and sandwich graded architectures.

Publications

Journal Articles

- S Baidya, S Ahmed, T Ahmed, **HR Paul**, MA Hasib, AM Billah, AK Debnath, "Experimental and Numerical Investigation of Flexural Energy Absorption in Octopus Suction-Cup-Inspired FDM-Printed Sandwich Structures." *Materials Today Communications*. (Under Review – revised manuscript submitted)
  SUMMARY: Studies octopus-sucker-inspired cellular cores for FDM-printed sandwich structures, comparing three core geometries (AV1–AV3) in flexural loading through experiments and finite element analysis, with a focus on energy absorption.

- **HR Paul**, MJ Rahman, SB Aesop, SMH Aronno, MA Hasib, AM Billah, AK Debnath, "Synergistic Effects of TiO_2/SiC Hybrid Ceramic Fillers on the Mechanical, Morphological and Moisture Resistance Behavior of Jute/Glass Fiber Epoxy Composites." (Manuscript in Preparation)
  SUMMARY: Examines five jute/glass fiber epoxy composites with varying TiO_2 (0–2 wt%) and SiC (0–3 wt%) filler content, using mechanical, SEM/EDS, FTIR and water absorption testing to link filler dispersion to performance.

Book Chapters

- MA Chowdhury, R Ahammad, **HR Paul**, S Roy, NUH Durjoy, AD Sarker, MM Rahman, "Additive Manufacturing and 3D Printing of Sandwich Structures," Chapter 11 in *Sandwich Structured Composites: Materials, Design and Applications*, Elsevier. (Under Review; expected March 2027).
  SUMMARY: Reviews additive manufacturing techniques (FFF, SLS, SLA, metal powder bed fusion) for producing sandwich panels with lattice, honeycomb and architected cores. Covers stiffness-to-weight efficiency, failure modes, interfacial bonding and print-induced anisotropy, along with applications, scalability and certification challenges, and emerging directions such as multi-material printing and functionally graded cores.

- MA Chowdhury, R Ahammad, **HR Paul**, S Roy, NUH Durjoy, AD Sarker, MM Rahman, "Sandwich Composites in Automotive, Rail, Aerospace, Unmanned Air Vehicle, and Marine Applications," Chapter 29 in *Sandwich Structured Composites: Materials, Design and Applications*, Elsevier. (Under Review; expected March 2027).
  SUMMARY: Analyzes the mechanical behavior, material architecture and manufacturing routes of sandwich composites across five transport sectors. Focuses on dominant failure mechanisms (face yielding, core shear, face-core debonding), crashworthiness and energy absorption, damage tolerance and inspection, and fire and environmental durability, with a view toward multifunctional architected cores and advanced non-destructive evaluation.

Academic Project

Compact Motorized Multi-Axis Stirring System – Individual Course Project (ME-3200):

- Designed and fabricated an automated stirring prototype integrating CAD modeling, PLA 3D printing, and Arduino Uno based motion control.
- Developed a three-axis mechanism for rotational mixing, vertical positioning, and angular alignment using DC gear motor and servo actuators.
- Implemented manual and automated control modes for sequential multi-container mixing using embedded electronics.

Technical Skills

- **CAD, Design & Simulation:** SolidWorks, AutoCAD, Ansys, Abaqus, CAD Modeling & Technical Drawing, Aerodynamic Simulation
- **Materials Testing & Characterization:** Tensile/Flexural Testing, SEM, EDS/EDX, FTIR, Shore D Hardness, Water Absorption Testing
- **Programming Language:** Python and C language.
- **Fabrication:** FDM/FFF 3D Printing, Filament Extrusion.
- **Software and Digital Tools:** Microsoft Office Suite, Google Workspace, WordPress, Technical Documentation.

Leadership & Technical Experience

IMechE KUET Student Chapter **Aug 2026 – Present**

Policy Analyst

- Plan and adapt the events the chapter organizes.
- Manage abstract submissions and participant registration for the SOfE (Speak Out for Engineers) heat using Google Forms and Excel.
- Prepare and maintain the formal documentation needed to run the chapter's activities.

CADers (Computer Aided Designers Club) **Mar 2024 – Present**

Quality Audit Officer (Aug 2026 - Present)

- Oversee the quality, compliance, and operational standards of club activities as part of the independent Audit Board, conducting periodic quality audits under the club constitution and reporting to the Faculty Advisor.

Chief Executive Officer (Jan 2025 – Aug 2026)

- Led a working committee of 40 members across 6 sub-teams.
- Organized separate SolidWorks and AutoCAD workshops, with an introduction to engineering design, for nearly 200 participants, supported by continuous evaluation and intra-club contests.
- Organized CAD contests, technical events, and guideline sessions for undergraduate students.
- Managed the club's social media, including a Facebook group of 11K+ members and a page with 7K+ followers.
- Collaborated with clubs and student organizations across Bangladesh on technical and community initiatives.

Assistant IT & Resource Officer (Mar 2024 – Jan 2025)

- Assisted instructors in workshops by supporting participants one-on-one, and gave feedback on participants' submitted homework.
- Produced YouTube tutorials by combining screen recordings and voice-over in Adobe Premiere Pro.
- Prepared sponsorship proposals and approached local tech companies; designed graphics for the club.

Team Kilo Flight (FSAE team)

Senior Executive – Body & Aerodynamics sub-team (Dec 2024-Aug 2026)

- Designed the hub and ran structural simulation to visualize stress and strain, then optimized the design.
- Assembled CAD models of the steering and braking systems.
- Interviewed new trainee members for the sub-team.

Trainee member – Body & Aerodynamics sub-team (Mar 2024-Dec 2024)

- Attended sessions to learn and practice airfoil and static structural simulation in Ansys Workbench, and studied topics for FSAE competition quizzes.

Teaching & Mentoring Experience

CADers, KUET **Dec 2024 – Mar 2026**

Instructor & Mentor (Computer aided design workshops)

- Delivered offline AutoCAD and SolidWorks workshops for undergraduate engineering students, with 40–50 participants per session and classes of nearly three hours.
- Prepared lecture content, presentations, and classwork and homework in advance, and demonstrated the software live while teaching.
- Mentored students preparing for national-level CAD contests and international certification exams.

10 Minute School (The largest online education platform in Bangladesh) **Aug 2023 – Jul 2025**

Junior Teacher - Doubt Solver (Contract)

- Resolved students' academic doubts in physics, chemistry, biology, and mathematics during and after live classes in 10 Minute School's academic programs, providing online guidance to students in grades 6–12, including O Level and A Level candidates.

Internship Experience

Youth School for Social Entrepreneurs (YSSE) (youth-led non-governmental organization) **Oct 2022 – Aug 2023**

Junior Associate (Jun – Aug 2023)

- Collected updates from Management Trainees and reported to the department head; coordinated with the YSSE President to request approval for sessions and campaigns.
- Interviewed intern applicants, published selected blogs on the WordPress site, and kept in contact with other departments.

Management Trainee (Mar – Jun 2023)

- Managed a group of content-writing interns: approved blog captions, assigned proofreading tasks, tracked deadlines, and hosted online meetings.

Intern-Content Writing Department (Oct 2022 – Mar 2023)

- Wrote blogs for youth readers that were published on the YSSE blog site and proofread colleagues' drafts before final approval.

Certifications

- **Certified SOLIDWORKS Professional (CSWP)**, Dassault Systèmes, 2024
- **Certified SOLIDWORKS Associate – Mechanical Design (CSWA)**, Dassault Systèmes, 2024
- **Technical Drawing & CAD Competition**, CADers, 2023
- **Professional Content Writing**, BrightSkills, 2022

Honors and Awards

- **2nd Runner Up, National CAD Contest – Ignition 2026**, organized by the Dept. of Mechanical Engineering, KUET (May 2026)
- **1st Runner Up, National CAD Contest – TECHNOMIZE 1.0**, organized by IEM Robotics & CAD Club, KUET (Nov 2024)
- **1st Runner Up, National CAD Contest – Calibration 1.0**, organized by the Dept. of Mechatronics Engineering, KUET (Feb 2024)
- **Technical Scholarship recipient**, Khulna University of Engineering & Technology (KUET): awarded for the 1st-year and 2nd-year academic sessions.

Volunteer and Extracurricular Activities

- **Volunteer, Hult Prize at KUET 2024** (Jan – Feb 2024): assisted in event coordination and operations during the competition.
- **Executive Committee Member, Josephite Math Club, Dhaka** (Aug 2019 – Dec 2021): assisted in conducting Math Olympiads and other competition segments for students up to higher secondary level at the Josephite Math Mania 2019.